2D CAD tools & sketching

National 5 · CAD & technology · Worksheet

GRAPHICS WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ name 2D CAD tools from a sequence of changes: line, trim, mirror, rotate, scale, fillet, chamfer
- ☐ explain the use of CAD libraries

I DO — Worked example

Read how a full-mark answer is built

A 2D CAD drawing of a shelf bracket is created in five stages: lines are drawn; overlapping lines are removed; the shape is copied as a reflection; the shape is turned 90°; the shape is enlarged.

1 State the name of the 2D CAD tool used at each stage. (5 marks)

Points that earn marks from the marking instructions: one ✓ per point

- ✓ Line
- ✓ Trim
- ✓ Mirror
- ✓ Rotate
- ✓ Scale (accept zoom only if the question allows)

WE DO — Guided practice

Use the starters and word bank to help you

An architect uses a CAD library when drawing floor plans.

2 Describe two advantages of using a CAD library. (2 marks)

Sentence starters

One … is …, which means …

Another … is …, so …

Word bank

sanitary · saving · accurate · library · standard · consistent · redrawn · symbols

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) A designer draws half a symmetrical shape. Which tool completes it?
☐ Rotate ☐ Mirror ☐ Fillet ☐ Zoom

b) Zoom changes...
☐ The size of the object ☐ The view only ☐ The scale on the drawing ☐ The line type

c) Which tool rounds a sharp corner?
☐ Chamfer ☐ Fillet ☐ Trim ☐ Extend

d) Which tool removes the overlapping parts of crossing lines?
☐ Mirror ☐ Trim ☐ Scale ☐ Copy

4 Key terms: write the letter of the matching description next to each term.

1. **Trim**	____	**A** Rounds a corner to a set radius.
2. **Mirror**	____	**B** Removes parts of lines up to where they meet other lines.
3. **Fillet (2D)**	____	**C** Creates a reflected copy about a line.

5 Explain what is meant by ‘Chamfer (2D)’. (2 marks)

6 Explain what is meant by ‘CAD library’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?